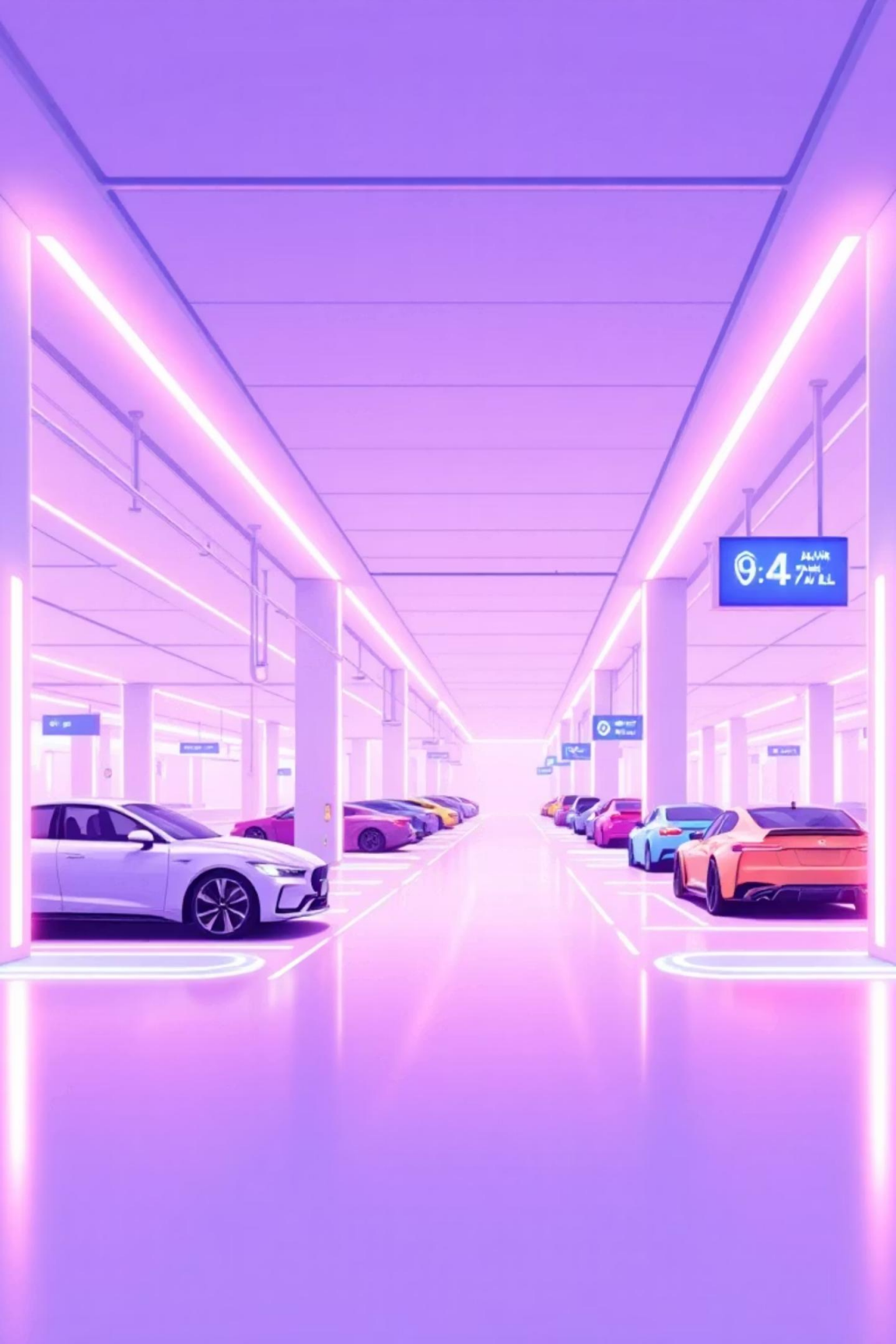




# Parking Management System

This mini-project demonstrates a robust parking management system built with Python and various data structures. It features efficient slot tracking, waitlist management, automated billing, and precise time tracking for vehicles.

## Presented By:

Rupesh Singh (USN ID: 25MCAR0017)

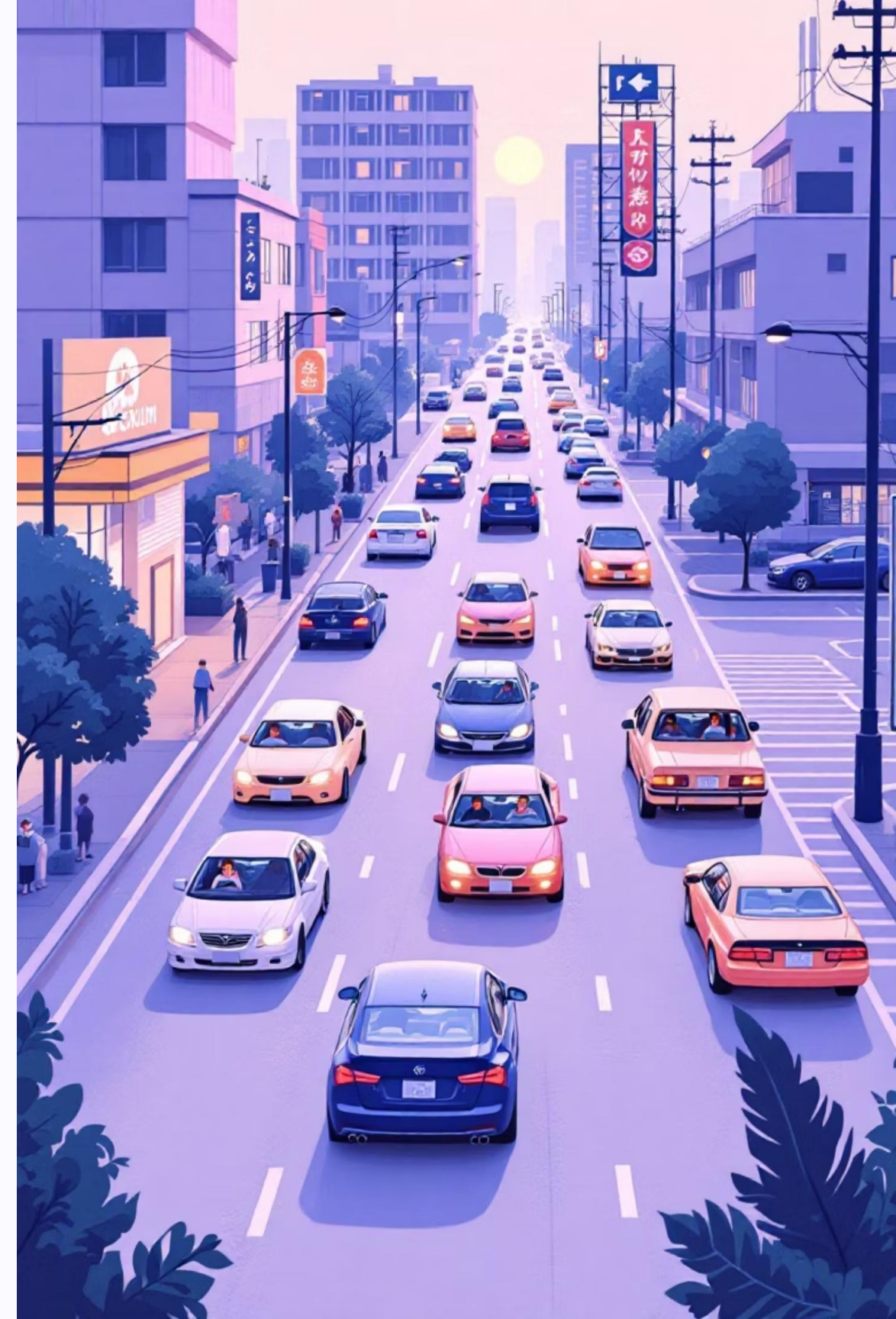
Vishwajeet Singh (USN ID: 25MCAR0219)

# Why a Parking Management System?

In today's urban environments, parking is a constant challenge. Limited spaces and the need for efficient management necessitate automated solutions. Our system addresses critical issues faced by traditional parking methods.

## Problem Statement

- Limited parking spaces require optimization.
- Manual tracking of vehicle entry/exit is inefficient.
- Need for automatic and accurate billing.
- Maintaining real-time slot availability is crucial.
- Handling overflow with a waitlist system.



# System Goals & Objectives

Our system is designed to provide comprehensive parking management, ensuring efficiency and user satisfaction. We aim to streamline operations and offer detailed reporting.

01

## Separate Parking

Manage car and bike parking independently.

02

## Slot Status

Maintain real-time status (Vacant / Filled).

03

## Automated Timestamps

Record entry and exit times automatically.

04

## Hourly Charges

Calculate parking fees based on duration.

05

## Waitlist Assignment

Automatically assign waitlisted vehicles to free slots.

06

## Status Report

Provide a full parking status report.





# Key Data Structures Used

The efficiency and reliability of our parking system are built upon carefully selected data structures, each serving a specific purpose to optimize performance.

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## Arrays (List)

Used to store parking slots for fast index-based access.

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## Queue (Deque)

Manages overflow with a FIFO approach, ensuring the first waiting vehicle gets the next free slot.

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## Dictionary

Enables  $O(1)$  lookup for parked vehicles using their registration number, ensuring quick retrieval.

---

## Class (OOP)

Organizes vehicle information, encapsulating data and behavior for better system management

# System Architecture Components

Our system's architecture is modular, comprising distinct components that work together to deliver a seamless parking experience. This design ensures scalability and maintainability.



# Vehicle Entry Process Flow

When a vehicle enters the parking facility, our system follows a precise sequence of steps to ensure efficient allocation and tracking, minimizing delays and maximizing space utilization.



## Identify Vehicle Type

System identifies if it's a car or a bike.



## Check Slot Status

Determines if a slot is Vacant or Filled for the respective vehicle type.



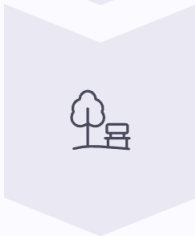
## Assign Slot

If free, assigns a slot and displays slot number, entry time, and status.



## Add to Waitlist

If parking is full, the vehicle is added to a waitlist.



## Show Available Count

Updates and displays the current count of available slots.

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# Dynamic Slot Status System

Our system provides real-time updates on parking slot availability, ensuring drivers can quickly identify vacant spots. This dynamic status helps in efficient parking management.



## Slot Array

Each parking type (car/bike) maintains its own slot array.



## Vacant Slot

Represented by 'None' in the array, indicating availability.



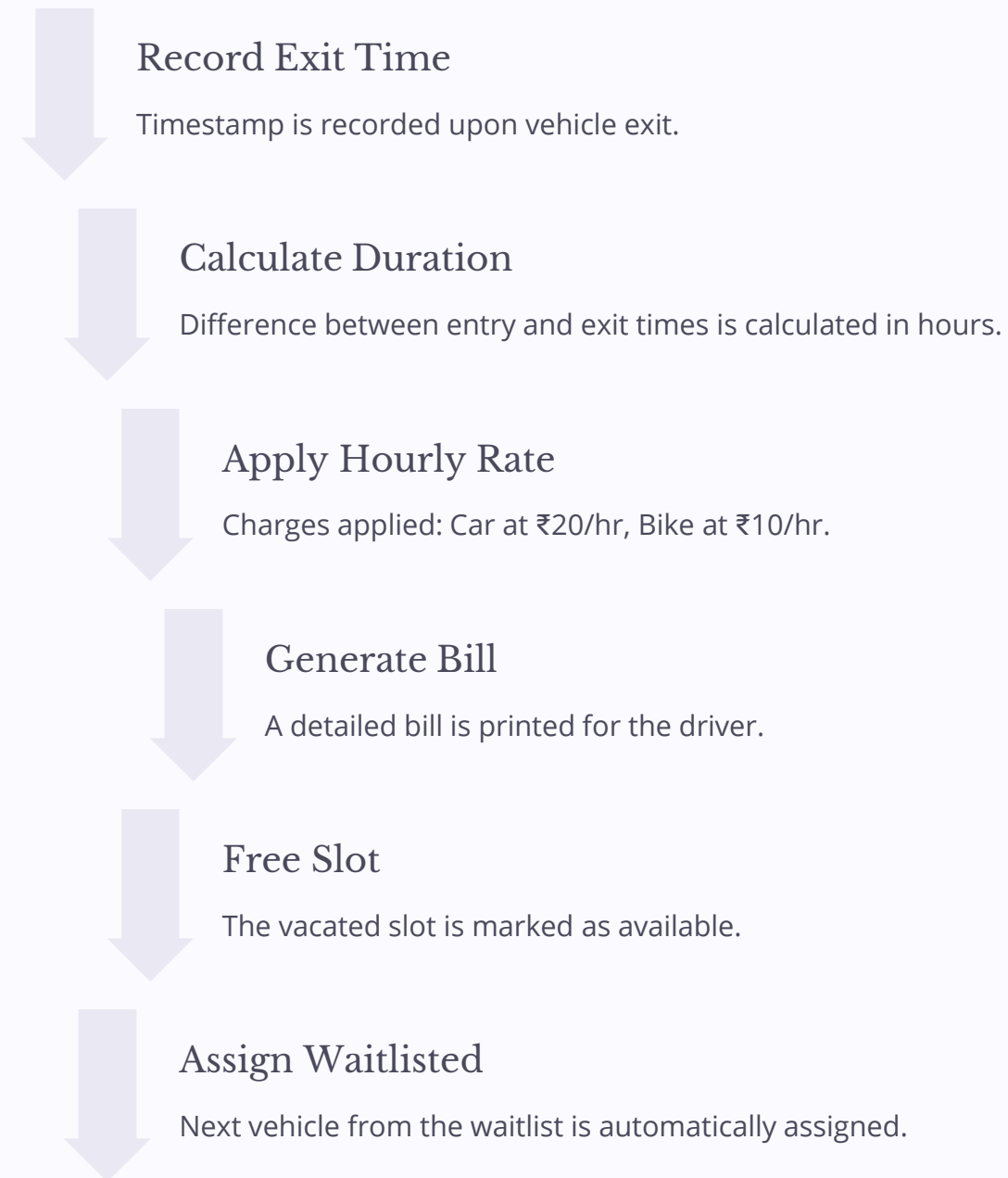
## Filled Slot

Contains a 'Vehicle Object', indicating an occupied slot.

The system clearly displays the status, for example: Slot 1: Vacant or Slot 1: Filled - MH12AB1234.

# Leaving Process & Automated Billing

The exit process is fully automated, from recording exit times to calculating charges and freeing up slots. This ensures a smooth departure and accurate billing for every vehicle.





# Billing Engine Details

Our billing engine is designed for transparency and accuracy, ensuring fair charges based on the actual parking duration and vehicle type. It's a core component of our automated system.



Car Rate  
Hourly rate for cars.



Bike Rate  
Hourly rate for bikes.



Time Tracking  
Precise calculation of parking duration.



Bill Generation  
Provides a physical record of charges.

The system converts total parking seconds into hours to apply the correct rate, ensuring no overcharging and clear accountability.

**PARKING** 

▲ **Vehicle**



↑ **\$.70₹**

H.AVWING



# Automated Parking System: Billing Example

Our system provides clear and concise billing for every vehicle, ensuring transparency and ease of use. Below is a sample bill generated by the system, detailing all relevant parking information and charges.

Vehicle No: MH12AB1234

Type: car

Entry Time: 2025-12-10 18:40:10

Exit Time: 2025-12-10 20:10:20

Hours Stayed: 2 hour(s)

Rate (/hr): ₹20

Total Bill: ₹40

# Efficient Waitlist Management

When the parking facility is full, our system intelligently manages incoming vehicles through a waitlist, ensuring fair and organized access. This feature optimizes space utilization and customer flow.

01

## Vehicle Added to Queue

Vehicles are added to a First-In, First-Out (FIFO) queue when the parking lot is at full capacity.

03

## System Updates

The system displays the assigned slot, a new entry time, and updated slot availability in real-time.

02

## Automatic Assignment

As soon as a slot becomes available, the next waitlisted vehicle is automatically assigned to it.



# System Overview

```
----- PARKING SYSTEM -----
```

1. Park Vehicle
2. Leave Vehicle
3. Show Full Status
4. Exit

```
-----
```

Enter choice: *1*

Enter Registration No: *KA02CK5555*

Enter Type (car/bike): *CAR*

Owner Name (optional): *Raj*

Vehicle parked successfully.

Type : car

Slot No : 1

Entry Time: 2025-12-10 22:27:13

Slot status (before -> after):

Slot 1: Vacant -> Slot 1: Filled

```
---- Car Slots ----
```

1. Filled - KA02CK5555 (Entry: 2025-12-10 22:27:13)
2. Filled - KA02cj0369 (Entry: 2025-12-10 22:28:22)
3. Filled - UP68CL0369 (Entry: 2025-12-10 22:28:42)
4. Filled - GJ15CZ9666 (Entry: 2025-12-10 22:29:12)
5. Filled - RJ06CX0636 (Entry: 2025-12-10 22:29:29)

```
---- Waitlists ----
```

Cars waitlist: CJ06153645

Bikes waitlist: (empty)

Available counts: Cars: 0, Bikes: 5

```
----- PARKING SYSTEM -----
```

1. Park Vehicle
2. Leave Vehicle
3. Show Full Status
4. Exit

```
-----
```

Enter choice:



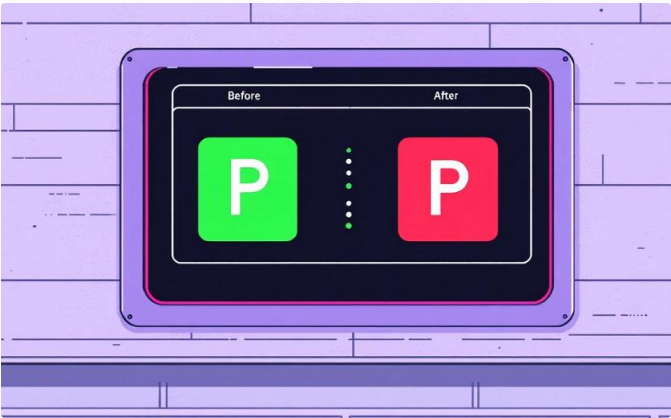
# Comprehensive System Demo

Our demonstration showcases the full capabilities of the automated parking system, from vehicle entry to exit, including advanced features like waitlist management and real-time status updates.



Parking a Vehicle

Seamless entry and automatic slot assignment.



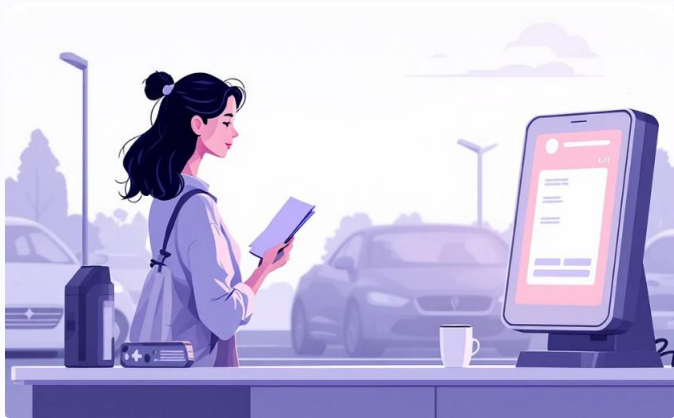
Slot Display Updates

Real-time changes in slot availability.



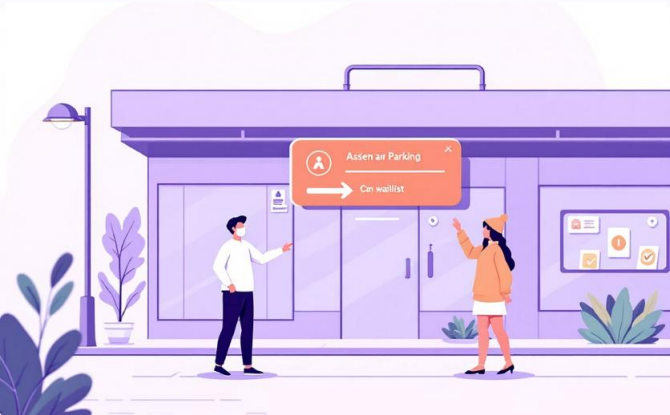
Availability Counts

Clear overview of occupied and vacant spaces.



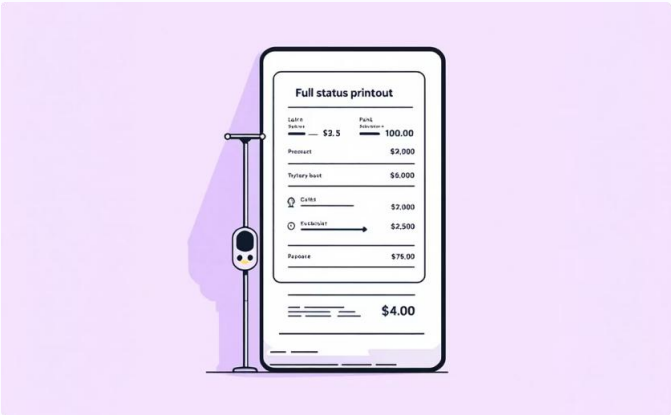
Leaving with Bill

Automated billing and payment process.



Waitlist Assignment

Efficient allocation for queued vehicles.



Full Status Printout

Comprehensive report of system operations.



# Future Enhancements & Conclusion

Our Parking Management System provides a robust foundation for efficient parking. We envision future enhancements to further improve user experience and operational intelligence.

## Potential Enhancements

- Mobile app integration for pre-booking.
- Integration with payment gateways.
- Real-time occupancy maps for users.
- Dynamic pricing based on demand.
- AI-powered analytics for parking optimization.

## Key Takeaways

- Efficient use of data structures.
- Automated and transparent processes.
- Scalable and modular architecture.
- Improved user experience and management.

